



(<https://learningcenter.hfsa.org/>).



My Courses (<https://learningcenter.hfsa.org/Users/Home.aspx>)
/ **2026 AHFTC Board Certification Review - OnDemand**
(<https://learningcenter.hfsa.org/Users/UserOnlineCourse.aspx?LearningActivityID=4f%2fOexzNsorKAF1lCFotvA%3d%3d>)
/ **2026 AHFTC Board Certification Question Bank**
(<https://learningcenter.hfsa.org/Users/UserOnlineCourse.aspx?LearningActivityID=YSpCYks4EUa7JCy700Rkkw%3d%3d>)
/ **Practice Exam - 1**

Correct Answer

Next Question

Practice Exam - 1

Question 21 of 117



A 43-year-old woman with dilated non-ischemic cardiomyopathy (LV 8.0 cm) with LV ejection fraction 10% presents with anorexia and mild elevation of liver chemistry tests (2X upper limit of normal). Examination reveals an estimated jugular venous pressure of 15 cm of water. Echo reveals stable LV size and systolic function, mild-to-moderate aortic regurgitation and new RV dilation and severe RV contractile dysfunction. She diureses poorly over the first 24 hours in response to intravenous milrinone and high dose IV diuretics. Creatinine rises to 1.4 mg/dl from 1.0 mg/dl. Lactate is 5 mmol/L. Blood pressure is 80/60/70 mm Hg. CT angiography reveals a sub-segmental pulmonary embolism in the left lower lung. She is referred for right heart catheterization which reveals:

- RA 22 mm Hg
- PA 63/36/49 mm Hg
- PCW 34 mm Hg
- CO 3.0 L/min
- CI 1.2 L/min/m²
- PVR 5.0 Wood units
- PA sat 26%

Which of the following would be the next best step to manage this patient?

A. Intra-aortic balloon pump insertion

 **B.** Ventriculo-arterial ECMO

C. Addition of sodium nitroprusside

D. Addition of intravenous norepinephrine

Commentary References

Testing point: The Board-certified AHFTC specialist should recognize which is the best treatment for cardiogenic shock according to clinical situation.

Answer: B. ventriculo-arterial ECMO.

This patient has severe cardiogenic shock despite inotropic support and requires aggressive measures to save her life. VA-ECMO could also be employed with concomitant LV venting strategies (not a choice here), but that can be done as the next step.

The other choices here would remain insufficient to salvage perfusion.

Answer C: The patient's LV cardiac power output, or $(\text{MAP} \times \text{CO}) / 451 = 0.46$, which is too low to be remedied with nitroprusside alone.

Answer D: Norepinephrine may boost blood pressure but at the expense of imposing more afterload.

Answer A: Due to very low cardiac power as well as the aortic regurgitation, intra-aortic balloon placement, or even Impella alone, would both be less optimal choices for managing her shock.

Powered by Oasis.



(<https://www.oasis-lms.com/>).